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The Area of a Long Skinny Interval in Two-dimensional Spacetime

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Submitted in partial fulfilment of the requirements for the degree of Master of Science of
Imperial College London, September 2017

Abstract

In this paper, the ‘long skinny interval’ problem is studied. The two endpoints of a causal diamond are joined by a timelike geodesic which has a proper length of τ . The work is done in two-dimensional Null Fermi Normal Coordinate system and the expression for the area of a long skinny interval is found. We first find the affine parameter along a timelike geodesic and then write the area in terms of τ . It turns out that the corrections of the area depend on the curvature along the geodesic, which is exactly what one should expect. The form that the volume might take in four-dimensional spacetime is also given.

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1 Introduction

It has been many years for people to find a theory of quantum gravity[1]. The idea of Planckian scale spacetime and causality discreteness, known as ‘causal sets’, gives an approach to quantum gravity [2, 3]. There is some evidence for the discreteness, such as that the lack of a short distance cutoff results in the infinities in general relativity and quantum field theory[4]. In addition, the ‘entanglement entropy’ of quantum fields including the black hole entropy makes the Planckian scale discreteness unavoidable[5–8].

A causal set is a set of points in spacetime where the Lorentz symmetry is not violated [9, 10], constructed by the order relation $\prec$. Suppose one has two elements in the spacetime, say x and y , then one has $x \prec y$ if x is in the past light-cone of y . The set of all elements z such that $x \prec z \prec y$ is known as a causal diamond or an Alexandrov open set.

Causal diamonds and their volumes are becoming increasingly important since they enable people to find many interesting properties of the spacetime, such as the conformal, topological and differential structures [11–13]. Let $V(x, y)$ be the n -volume of a causal diamond C between x and y in n -dimensional spacetime, then the volume is [11]

$$V(x, y) = \int_C \sqrt{|g|} d^n z. \quad (1)$$

In particular, in four-dimensional Minkowski spacetime, ’t Hooft’s formula gives [14]

$$V(x, y) = \frac{\pi}{24} \tau^4, \quad (2)$$

where τ is the proper time of a unique timelike geodesic between x and y . In curved spacetime, the correction in terms of τ together with the curvature at the centre of the causal diamond is given in [11] and [15] where x and y are sufficiently close. To understand the deep structure of causal sets better, it is helpful for people to study

causal diamonds [2, 11]. The research on the scalar D'Alembertian can be a persuasive example.

1.1 The Continuum Scalar D'Alembertian

Though the theory of causal sets plays an important role in fundamental physics, it has a significant feature: the coexistence of spacetime discreteness and Lorentz symmetry will result in the loss of locality [9, 16, 17]. However, it has been proved that in two-dimensional Minkowski spacetime, one can solve this problem with the scalar D'Alembertian operator [3, 18]. Furthermore, this approach also applies when extended to four-dimensional Minkowski spacetime [19, 20] or other various dimensions [21, 22].

Let us now consider a causal set $\mathcal{C}$ and an element $x \in \mathcal{C}$, then we have the order interval defined as $I(x, y) := \{z \in \mathcal{C} \mid y \preceq z \preceq x\}$ [19, 20]. Local finiteness means that the cardinality of any order interval should be finite, and a causal set is a locally finite partial order. Then let $n(x, y) = |I(x, y)| - 2$, where $|\cdot|$ is the cardinality of a set.

Assume ϕ is a scalar field such that $\phi : \mathcal{C} \rightarrow \mathbb{R}$, then according to the definition of a discrete D'Alembertian B [19, 20], one has

$$B\phi(x) := \frac{4}{\sqrt{6}l^2}[-\phi(x) + \left(\sum_{y \in L_1} -9 \sum_{y \in L_2} + 16 \sum_{y \in L_3} - 8 \sum_{y \in L_4}\right) \phi(y)], \quad (3)$$

where $L_i = \{y \in \mathcal{C} \mid y \prec x, \text{ and } n(x, y) = i - 1\}$.

A Poisson process called sprinkling can generate a causal set from a Lorentzian manifold $(\mathcal{M}, g)$ [19, 20]. During sprinkling, one selects elements in $\mathcal{M}$ with density ρ , so the expected number of points in a region is ρV , with V being the spacetime volume. The sprinkled points are then the elements of this causal set, ordered by the

causal order of the manifold. We choose the Planckian discreteness length scale to be l and $\rho = l^{-4}$.

With ϕ being the real test field of compact support on $\mathcal{M}$ and $x \in \mathcal{M}$ being fixed, the expectation value of $B\phi(x)$ during the sprinkling is given by [19, 20]

$$\bar{B}\phi(x) = \mathbb{E}(B\phi(x)) = \frac{4\sqrt{\rho}}{\sqrt{6}}[-\phi(x) + \rho \int_{y \in J^-(x)} d^4y \sqrt{-g} \phi(y) e^{-\xi} (1 - 9\xi + 8\xi^2 - \frac{4}{3}\xi^3)], \quad (4)$$

where $J^-(x)$ is the causal past of point x , and $\xi = \rho V(y)$, while $V(y)$ is the spacetime volume of the causal interval between x and y . It has been shown that in various dimensions, as l goes to zero, i.e. ρ is large enough, d'Alembertians will reach a local limit [19–22]

$$\lim_{\rho \rightarrow \infty} \bar{B}\phi(x) = (\square - \frac{1}{2}R(x))\phi(x). \quad (5)$$

From Eqn.(4) can see the integral is suppressed when ξ is large, i.e. when the volume $V(y)$ is larger than a few Planck volumes. But the suppression will failed to be provided when y is close to the light-cone of x and ξ is small. As a result, one is not able to find a value which is a local quantity at x approximately from the exponential factor.

Eqn.(4) also indicates that the integral cannot be computed unless one knows how $V(y)$ behaves for every point $y \in J^-(x)$. One can quote the formula given by the authors of [11, 15] when y is close to x , where $V(y)$ is expanded in Riemann Normal Coordinates(RNC). However, for y in the region close the the boundary of $J^-(x)$ (as shown in Figure.1), the behaviour of $V(y)$ for this interval has never been studied so far.

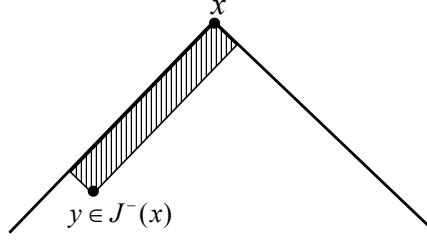


Figure 1: Every point $y \in J^-(x)$ should be considered in calculation. When y is in the region near the boundary of $J^-(x)$, the volume is so small that the integral will not be suppressed. An interval between x and y where y is close to the light-cone of x is called a long skinny interval.

Let us define N to be a Riemann normal neighbourhood of x with RNC $\{y^\mu\}$ and the origin x is the centre. We also define spatial polar coordinates: $r = \sqrt{\sum_{i=1}^3 (y^i)^2}$, θ and φ [20] as well. The radial null coordinates are defined as well: $u = \frac{1}{\sqrt{2}}(-y^0 - r)$ and $v = \frac{1}{\sqrt{2}}(-y^0 + r)$ in N where u and v increase into the past.

Then let $LC = \partial J^-(x) \cap \text{Support}(\phi)$, where $\text{support}(\phi)$ is the closure where ϕ is non-vanishing. Suppose that every point from LC is on a unique past directed null geodesic from x [20]. Furthermore, we define Null Gaussian Normal Coordinates (NGNC) [23] $\{U, V, \theta, \varphi\}$ in N_{LC} , which is neighbourhood of $\partial J^-(x)$.

In Null Fermi Normal Coordinates, y has coordinates $(x^+ = V, x^- = U, x^1 = 0, x^2 = 0)$. We will give the construction of Null Fermi Normal Coordinates later. When U is small enough, the volume can take the form [20]

$$V(y) = U^2 f_0(V, \theta, \varphi) + U^3 G(V, U, \theta, \varphi), \quad (6)$$

where f_0 is an increasing function of V and G is a continuous function. Note that V refers to the coordinate and $V(y)$ is the volume.

Now we would like to split the region of integration into three parts, W_1 , W_2 and

W_3 [20]. The near region is W_1 and the definition is

$$W_1 := \{y \in N \mid 0 \leq u \leq v \leq a\}, \quad (7)$$

where $a > 0$ and W_1 is approximately flat. The next is the region “down the light-cone”

$$W_2 := \{y \in N_{LC} \mid 0 < a'(\theta, \varphi) \leq V \leq L, \quad \text{and} \quad 0 \leq U \leq \frac{b^2}{\sqrt{f_0(V, \theta, \varphi)}}\}, \quad (8)$$

and L is big enough to cover the whole LC outside W_1 . Here a is the value of RNC v and a' is the value of NFNC V . Finally, the definition of the “deep chronological past” W_3 :

$$W_3 := (\text{Supp}(\phi) \cap J^-(x)) \setminus (W_1 \cup W_2) \quad (9)$$

and it is bounded away from LC .

In curved spacetime, $u \neq U$ and $v \neq V$, so there is a mis-alignment between the boundaries of W_1 and W_2 regardless of the no matter what choice is made for a' [20]. If the normal neighbourhood of N is approximately flat, then $u \simeq U$ and $v \simeq V$, and the mismatch can be decreased by taking a to be zero. In [20], $a \neq a'$ while they are almost the same, and here we quote the Figure.2 from [20].

Let us define an operator:

$$\hat{\mathcal{O}} := \frac{4}{3}(H + \frac{1}{2})(H + 1)(H + \frac{3}{2}) \quad (10)$$

$$= 1 + 9H_1 + 8H_2 + \frac{4}{3}H_3, \quad (11)$$

where

$$H_n := \rho^n \frac{\partial^n}{\partial \rho^n} \quad \text{and} \quad H := H_1. \quad (12)$$

Note that

$$\hat{\mathcal{O}}e^{-\xi} = e^{-\xi} \left(1 - 9\xi + 8\xi^2 - \frac{4}{3}\xi^3 \right). \quad (13)$$

In addition, we also define [20]

$$I_i = \hat{\mathcal{O}} \int_{W_i} d^4y \sqrt{-g(y)} \phi(y) e^{-\rho V(y)}, \quad (14)$$

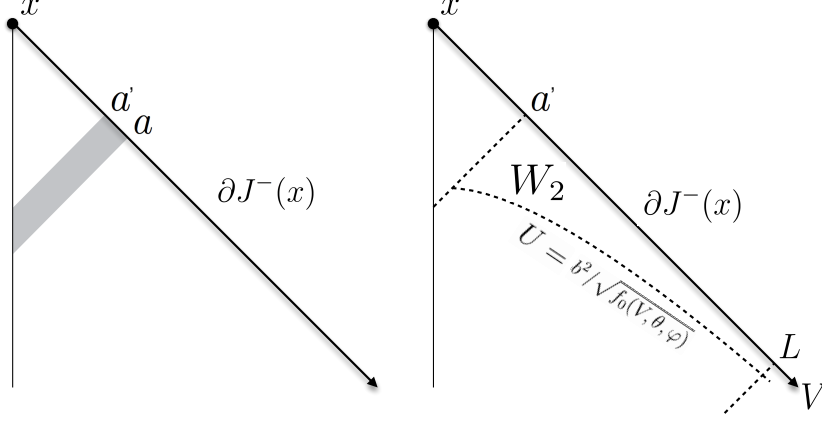


Figure 2: The left diagram illustrates the overlap between W_1 and W_2 , which is shaded. The integration region W_2 is showed in the right diagram.(Quoted from [20])

for $i = 1, 2, 3$ so that

$$\bar{B}\phi(x) := \frac{4\sqrt{\rho}}{\sqrt{6}} \left[-\phi(x) + \rho(I_1 + I_2 + I_3) \right]. \quad (15)$$

In W_2 , if one works in NGNC and makes U small enough, then one has [20]

$$V(y) = U^2 f_0(V, \theta, \varphi) + U^3 f_1(V, \theta, \varphi) + U^4 f_2(V, \theta, \varphi) + U^5 F(y), \quad (16)$$

and

$$\begin{aligned} e^{-\rho V(y)} &= e^{-\rho U^2 f_0} \left(1 - \rho U^3 (f_1 + U f_2 + U^2 F) + \frac{1}{2} \rho^2 U^6 (f_1 + U f_2 + U^2 F)^2 \right. \\ &\quad \left. + \sum_{k=3}^{\infty} \frac{(-\rho)^k}{k!} U^{3k} (f_1 + U f_2 + U^2 F)^k \right). \end{aligned} \quad (17)$$

Furthermore, $\sqrt{-g(y)}$ and $\phi(y)$ are also expanded as:

$$\sqrt{-g(y)} = h_0(V, \theta, \varphi) + U h_1(V, \theta, \varphi) + U^2 h_2(V, \theta, \varphi) + U^3 H(y), \quad (18)$$

$$\phi(y) = \phi|_{U=0} + U \phi_{,U}|_{U=0} + \frac{1}{2} U^2 \phi_{,UU}|_{U=0} + U^3 \Phi(y), \quad (19)$$

where the functions F , H and Φ are continuous. One needs to calculate

$$I_2 = \hat{\mathcal{O}} \int d\Omega_2 \int_{a'}^L dV \int_0^{\frac{b^2}{\sqrt{f_0(V, \theta, \varphi)}}} dU \sqrt{-g(y)} \phi(y) e^{-\rho V(y)}. \quad (20)$$

However, as one lacks an explicit expression for the volume ‘down the light-cone’ in W_2 , where the y is close to the light-cone of x , one can only find the integrals in terms of some unknown functions, for instance, $f_0(V, \theta, \phi)$ [20]. Hence computing the volume of a long skinny interval is essential for one to know the scalar D’Alembertian better.

1.2 Long Skinny Intervals

In this paper, we would like to investigate the ‘long skinny causal interval problem’: suppose there is a unique future-pointing timelike geodesic from x to y , with τ being the proper time along it, as shown in Fig.3. On the other hand, starting from point x with tangent vector T^μ , γ is a unique future-pointing null geodesic.

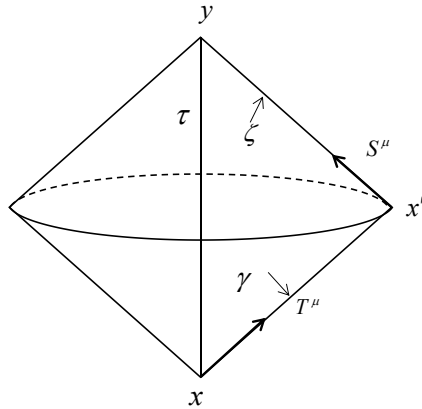


Figure 3: A causal diamond between x and y . The unique timelike geodesic between x and y has a proper time τ . The point x' is fixed and T^μ is the tangent vector along γ , which is lightlike. Another null geodesic ζ is from x' to y , with tangent vector S^μ .

Assume that there are future-pointing null geodesics ζ starting from points on γ with tangent vectors S^μ . Note that S^μ is everywhere on γ and is parallel transported along it. Let x' be an arbitrary point on γ . Here we normalise T^μ and S^μ , such that

at x'

$$g_{\mu\nu}(x')T^\mu S^\nu = -1. \quad (21)$$

Before making τ to be infinitely small, fix x' and consider a limit along ζ where $y \rightarrow x'$. Working in Null Fermi Normal Coordinates (NFNC) based on the affine coordinates along γ and ζ , we assume that $R_{abcd}\tau^2 \ll 1$, where R_{abcd} are Riemann tensors in NFNC.

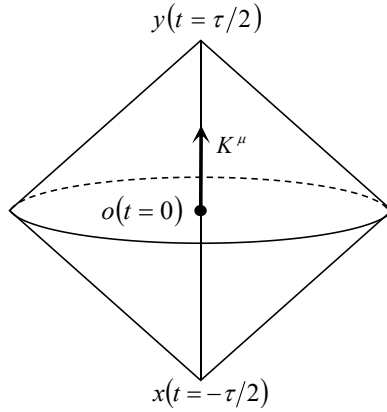


Figure 4: A ‘short fat interval’ between x and y . The points x and y are joined by a unique timelike geodesic, the affine parameter along which is t . Usually we choose x to be at $t = -\tau/2$ and y to be at $t = \tau/2$, so that the geodesic has a proper length of τ and the mid point o is at $t = 0$. The tangent vector along this geodesic is K^μ . If x and y close to each other along the geodesic then the evaluation of the volume will depend on the curvature at the mid point o .

Before we get started, it can be helpful to review ‘short fat intervals’ at the very beginning, which is plotted in Fig.4. The volume of a ‘short fat interval’ is computed in [11] and [15], where $y \rightarrow x$ along a unique timelike geodesic connecting them inside the diamond. Suppose the mid point of the timelike geodesic is o , then the metric for Riemman normal coordiantes is [11]

$$g_{\mu\nu} = \eta_{\mu\nu} - \frac{1}{3}R_{\mu\rho\nu\sigma}(o)x^\sigma x^\rho + \dots \quad (22)$$

With τ being the proper length of the geodesic, the authors of [11] and [15] found that

$$V(x, y) = \frac{\pi}{24} \tau^4 (1 + a\tau^2 R(o) + bR_{\mu\nu}(o)K^\mu K^\nu + \dots), \quad (23)$$

where K^μ is the tangent vector along the timelike geodesic. With the normalisation

$$g_{\mu\nu}(o)K^\mu K^\nu = -\tau^2, \quad (24)$$

the Eqn.(23) can be written as [11]

$$V(x, y) = \frac{\pi}{24} \tau^4 (1 + a\tau^2 R(o) + b\tau^2 R_{00}(o) + \dots), \quad (25)$$

where R_{00} is the time-time component of the Ricci tensor. Vector T^μ can also be expressed by Synge's world function $\Omega(x, y)$ [11, 24] which satisfies

$$g^{\mu\nu} \partial_\mu \Omega \partial_\nu \Omega = -4\Omega. \quad (26)$$

So one has [11]

$$T^\mu = \frac{1}{2} g^{\mu\nu} \partial_\nu \Omega. \quad (27)$$

One can define

$$\tau = \sqrt{\Omega(x, y)}, \quad (28)$$

and write Eqn.(26) as

$$g^{\mu\nu} \partial_\mu \partial_\nu \tau = 1, \quad (29)$$

which is known as the Hamilton-Jacobi equation.

To fix the coefficients a and b , one needs to calculate the volume of a causal diamond in some explicit metrics first, then compare the results with Eqn.(25)[11].

For example, consider the Einstein Static Universe where

$$ds^2 = -dt^2 + d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2). \quad (30)$$

One can choose $x = (-\tau/2, 0, 0, 0)$ and $y = (\tau/2, 0, 0, 0)$ [11]. One also has $R = 6$ and $R_{00}(o) = 0$. Therefore,

$$\begin{aligned}
V(x, y) &= 8\pi \int_0^{\frac{1}{2}\tau} dt \left(\int_0^{\frac{1}{2}\tau-t} \sin^2 \chi d\chi \right) \\
&= \frac{1}{2}\pi \left(\cos^2 \frac{\tau}{2} + \frac{\tau^2}{4} - 1 \right) \\
&= \frac{\pi}{24} \tau^2 \left(1 - \frac{\tau^2}{30} + \dots \right)
\end{aligned} \tag{31}$$

at small τ [11]. Hence $a = -1/180$.

Similarly, the metric for de-Sitter spacetime is

$$ds^2 = -dt^2 + \cosh^2 t \left(d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2) \right). \tag{32}$$

With the same choice for x and y , one finds $R = 12$ and $R_{00}(o) = -3$. Then [11]

$$V(x, y) = \frac{\pi}{24} \tau^4 \left(1 - \frac{1}{6} \tau^2 + \dots \right). \tag{33}$$

Thus one can fix $b = 1/30$.

We are now interested in the behaviour of the spacetime volume of a long skinny causal intervals where y is sufficiently close to the light-cone of x , and whether they are in any sense close to ‘flat’. Though we work in two-dimensional spacetime and what we find is actually the ‘area’ of the causal diamond, we will use the word ‘volume’ instead, since sometimes we refer to other dimensions.

Usually, in a two-dimensional spacetime with coordinate system $\{u, v\}$, supposing $x = (0, 0)$ and $y = (U, V)$, one has $\tau^2 = UV$ [25] when both U and V go to zero (called a ‘coincident limit’) and thus can express the area in terms of the proper length τ according to this relation. However, what we are considering in this paper is the situations where V is fixed while U goes to zero, which is not a ‘coincident limit’. As no one has ever studied the expression for τ by U and V in this situation for the time

being, we have to find it on our own originally.

The paper goes as follows: Firstly, we will construct the Null Fermi Normal Coordinates for two-dimensional spacetime. Secondly, we will solve the geodesic equation and obtain the expression of the proper length between x and y for a ‘non-coincident limit’. After that, we will give the relation between the two-volume for a long skinny interval and its proper length. Finally the results will be extended to four-dimensional spacetime.

2 Two-dimensional Metric in NFNC

2.1 The Construction of NFNC

Suppose there is a Riemann manifold, then we can construct coordinates near an arbitrary curve in this manifold [26, 27]. At the beginning, one should make the Christoffel symbols on this curve to be zero[27]. But it has been shown that some Christoffel symbols can be non-vanishing [24], leaving the metric rectangular and such coordinates are called Fermi Coordinates. If the curve is a geodesic, and the Christoffel symbols vanish along it, then the introduced coordinate are Fermi Normal Coordinates. Furthermore, the Fermi Coordinates for a null geodesic are known as Null Fermi Normal Coordinates(NFNC), in which we would like to study the long skinny interval problems.

The construction of Fermi normal coordinates has been shown in detail in [26, 28–30]. We would briefly introduce how to construct NFNC $\{x^+, x^-, x^a\}$ in a generic way, where $x^a = x^3, x^4, \dots, x^d$ in d-dimensional spacetime. First of all, let us choose a null geodesic γ on the manifold, the affine parameter along which is x^+ . Then there exist sets of geodesics emanating from γ , whose tangent vectors span the tangent space $\{x^-, x^a\}$. Note that $\{x^+, x^-, x^a\} = \{x^+, x^{\bar{a}}\}$, and the expansion of a general metric in NFNC is [20, 28]

$$\begin{aligned}
ds^2 = & -2dx^+dx^- + \delta_{ab}dx^adx^b \\
& -R_{+\bar{a}+\bar{b}} x^{\bar{a}}x^{\bar{b}}(dx^+)^2 - \frac{4}{3}R_{+\bar{b}\bar{a}\bar{c}}x^{\bar{b}}x^{\bar{c}}(dx^+dx^{\bar{a}}) - \frac{1}{3}R_{\bar{a}\bar{c}\bar{b}\bar{d}}x^{\bar{c}}x^{\bar{d}}(dx^{\bar{a}}dx^{\bar{b}}) \\
& -\frac{1}{3}R_{+\bar{a}+\bar{b};\bar{c}} x^{\bar{a}}x^{\bar{b}}x^{\bar{c}}(dx^+)^2 - \frac{1}{4}R_{+\bar{b}\bar{a}\bar{c};\bar{d}} x^{\bar{b}}x^{\bar{c}}x^{\bar{d}}(dx^+dx^{\bar{a}}) - \frac{1}{6}R_{\bar{a}\bar{c}\bar{b}\bar{d};\bar{e}} x^{\bar{c}}x^{\bar{d}}x^{\bar{e}}(dx^{\bar{a}}dx^{\bar{b}}) \\
& +(\frac{1}{3}R_{+\bar{a}A\bar{b}}R^A_{\bar{c}+\bar{d}} - \frac{1}{12}R_{+\bar{a}+\bar{b};\bar{c}\bar{d}}) x^{\bar{a}}x^{\bar{b}}x^{\bar{c}}x^{\bar{d}}(dx^+)^2 \\
& +(\frac{2}{15}R_{+\bar{b}A\bar{c}}R^A_{\bar{d}\bar{a}\bar{e}} - \frac{1}{15}R_{+\bar{b}\bar{a}\bar{c};\bar{d}\bar{e}}) x^{\bar{b}}x^{\bar{c}}x^{\bar{d}}x^{\bar{e}}(dx^+dx^{\bar{a}}) \\
& +(\frac{2}{45}R_{A\bar{c}\bar{a}\bar{d}}R^A_{\bar{e}\bar{b}\bar{f}} - \frac{1}{20}R_{\bar{a}\bar{f}\bar{b}\bar{c};\bar{d}\bar{e}}) x^{\bar{c}}x^{\bar{d}}x^{\bar{e}}x^{\bar{f}}(dx^{\bar{a}}dx^{\bar{b}}) \\
& +\mathcal{O}(x^{\bar{a}}x^{\bar{b}}x^{\bar{c}}x^{\bar{d}}x^{\bar{e}}),
\end{aligned} \tag{34}$$

where $R_{\mu\nu\rho\sigma}$ are the Riemann tensors in this metric and they are functions of x^+ only, i.e. $R_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma}(x^+)$.

2.2 Two-dimensional Spacetime in NFNC

As we know, every two-dimensional spacetime is conformally flat (see Appendix C for a quick derivation). Now consider a most general form of a two-dimensional metric in null-coordinates, where

$$ds^2 = 2a(u, v)dudv, \quad (35)$$

and generally, one has

$$a(u, v) = a_1(v) + a_2(v)u^2 + \dots \quad (36)$$

But here we choose $a_1(v) = -1$ so that we can make $x^+ = v$, which will be shown from coming calculation. So

$$a(u, v) = -1 + a(v)u + \mathcal{O}(u^2). \quad (37)$$

According to Eqn.(34), the expansion of a two-dimensional metric in NFNC $\{x^+, x^-, x^a = 0\}$ to the second order is

$$ds^2 = -2dx^+dx^- - (x^-)^2 R_{+-+ -}(dx^+)^2 + \mathcal{O}((x^-)^3), \quad (38)$$

where x^+ is the affine parameter along the null geodesic γ and x^- is the affine parameter along ζ , which are the geodesics orthogonal to γ . From Eqn.(38) one can see we are working in an expansion of x^- . That x^- is small and that u is small are consistent, which will be shown later. As a result, one is also working in an expansion of small u .

Following the steps in [26, 28, 31], one needs first to get the affine parameter along a null geodesic from origin and then set the parameter to be the coordinate x^+ . Hence

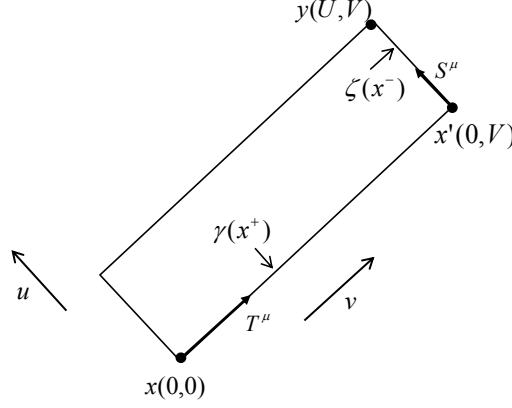


Figure 5: A causal diamond in two-dimensional null coordinates. The affine parameter along a unique null geodesic γ in this metric is set to be x^+ in NFNC. Then the affine parameter of the set of geodesics with tangent vector S^μ such that along γ one has $g_{\mu\nu}T^\mu S^\nu = -1$ is x^- , where T^μ is the tangent vectors of γ .

to give the expansion of metric (35) in Null Fermi Normal Coordinates, one has to solve the geodesic equations for x^+ along the null geodesic γ .

The geodesic equations take the form

$$\left(\frac{d}{dx^+}\right)^2 u = -\frac{a(v)}{-1 + a(v)u} \left(\frac{du}{dx^+}\right)^2, \quad (39)$$

$$\left(\frac{d}{dx^+}\right)^2 v = -\frac{a'(v)u}{-1 + a(v)u} \left(\frac{dv}{dx^+}\right)^2 = -\frac{\tilde{a}'(x^+)u}{-1 + \tilde{a}(x^+)u} \left(\frac{dv}{dx^+}\right), \quad (40)$$

where $\tilde{a}(x^+) = a(v(x^+))$, and chain rule has been used in the second equation. Since γ is null, the tangent vector T^μ along γ should satisfy the null condition:

$$g_{\mu\nu}T^\mu T^\nu = (-1 + a(v)u) \left(\frac{du}{dx^+}\right) \left(\frac{dv}{dx^+}\right) = 0. \quad (41)$$

Along the geodesic γ , $du/dx^+ = 0$, then from Eqn.(40)one finds

$$dv|_\gamma = \frac{C_p}{1 - \tilde{a}(x^+)u} dx^+ = dx^+, \quad (42)$$

since $u = 0$ along γ and C_p is fixed to be 1.

Now consider a set of geodesics ζ orthogonal to γ , with affine parameter x^- and tangent vector S^μ . Similarly, along ζ , $dv/dx^- = 0$ and the geodesic equation is (see Appendix A for more general cases)

$$\left(\frac{d}{dx^-}\right)^2 u = -\frac{a(v)}{-1 + a(v)u} \left(\frac{du}{dx^-}\right)^2. \quad (43)$$

It is easy to show that

$$\begin{aligned} 0 &= a(u, v) \left(\frac{d}{dx^-}\right)^2 u + \frac{da(u, v)}{du} \left(\frac{du}{dx^-}\right)^2 \\ &= \frac{d}{dx^-} \left(a(u, v) \frac{du}{dx^-} \right). \end{aligned} \quad (44)$$

This vindicates,

$$a(u, V) \frac{du}{dx^-} = \text{const.} \quad (45)$$

Can assume that

$$x^- = u + u^2 f(v) + \mathcal{O}(u^3), \quad (46)$$

and substitute into the metric (38) to the first order of u such that

$$ds^2 = -2dx^+ dx^- + \mathcal{O}(u^2) = 2a(u, v) du dv. \quad (47)$$

Hence we find

$$f(v) = -\frac{1}{2}a(v). \quad (48)$$

Thus,

$$x^+ = v, \quad (49)$$

$$x^- = u - \frac{1}{2}a(v)u^2. \quad (50)$$

In addition, one can assume that

$$u = mx^- + n(x^-)^2 + \mathcal{O}((x^-)^3). \quad (51)$$

Substituting Eqn.(51) into Eqn.(50), one obtains

$$x^- = mx^- + n(x^-)^2 - \frac{1}{2}a(v)(x^-)^2 + \mathcal{O}((x^-)^3). \quad (52)$$

Then one finds

$$m = 1, \quad (53)$$

$$n = \frac{1}{2}a(v), \quad (54)$$

so

$$u = x^- + \frac{1}{2}a(v)(x^-)^2 + \mathcal{O}((x^-)^3), \quad (55)$$

which shows that when u is small then x^- is small and vice versa.

Now the only thing that left is the relation between R_{+--+} and $a(u, v)$. To solve this problem, one can compute the Ricci scalar R for both two metrics and find the relation, as Ricci scalar is invariant.

As we know,

$$R = g^{\mu\nu} R_{\mu\nu} = g^{\mu\nu} R^\alpha{}_{\mu\alpha\nu}. \quad (56)$$

For metric (35), one has

$$g^{uv} = g^{vu} = \frac{1}{a(uv)} = \frac{1}{-1 + a(v)u}, \quad (57)$$

and the non-vanishing Ricci tensors are

$$R_{uv} = R_{vu} = \frac{a(v)ua'(v)}{a(u, v)^2} - \frac{a'(v)}{a(u, v)}. \quad (58)$$

So the Ricci scalar for metric (35) is

$$R|_{2d} = \frac{2a(v)a'(v)u}{a(u, v)^3} - \frac{2a'(v)}{a(u, v)^2} = -2a'(v) + \mathcal{O}(u) \quad (59)$$

at small u . Then for a metric in NFNC (38), one can find

$$g^{+-} = g^{-+} = -1, \quad (60)$$

$$g^{--} = (x^-)^2 R_{+--+}(x^+). \quad (61)$$

The non-zero components for the Ricci Tensors are

$$R_{++} = (x^-)^2 (R_{+--+}(x^+))^2, \quad (62)$$

$$R_{+-} = R_{-+} = R_{+--+}(x^+). \quad (63)$$

So for metric in NFNC, the Ricci scalar takes the form

$$R|_{NFNC} = -2R_{+--+}. \quad (64)$$

Since Ricci scalar is invariant, we have

$$R|_{2d} = R|_{NFNC} \Leftrightarrow -2a'(v) = -2R_{+--+}. \quad (65)$$

Obviously,

$$R_{+--+}(x^+) = a'(v). \quad (66)$$

Substituting the results into Eqn. (38) gives the expansion of metric (35) in NFNC:

$$ds^2 = 2a(u, v)dudv + u^2a'(v)dv^2 + \mathcal{O}(u^3). \quad (67)$$

3 The Two-volume of a Long Skinny Interval

To find the two-volume of a long skinny interval, we would like to first get the affine parameter τ for metric (38) along a timelike geodesic. We denote R_{+-+-} by $\mathcal{R}$ for convenience. Here y is set to be (U_F, V_F) in $\{x^-, x^+\}$ coordinates (see Fig.6 for details). The geodesic equations for τ read

$$\frac{d^2 x^+}{d\tau^2} - x^- \mathcal{R} \left(\frac{dx^+}{d\tau} \right)^2 = 0, \quad (68)$$

$$\frac{d^2 x^-}{d\tau^2} + \frac{1}{2}(x^-)^2 (2x^- \mathcal{R}^2 + \frac{d\mathcal{R}}{dx^+}) \left(\frac{dx^+}{d\tau} \right)^2 + 2x^- \mathcal{R} \frac{dx^+}{d\tau} \frac{dx^-}{d\tau} = 0. \quad (69)$$

We would like to obtain the relationship between x^+ and x^- before we find the affine parameter τ . Then we can then express τ as a function only of x^+ , which makes the calculation easier.

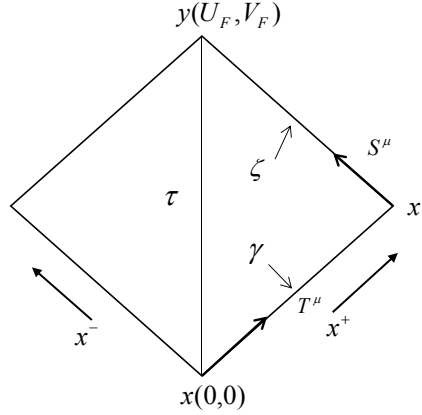


Figure 6: A causal diamond in 2-dimensional NFNC. Point x is set to be origin, and y is (U_F, V_F) . The null geodesic γ is unique, and x^+ is the affine parameter along γ . There is a set of geodesics ζ orthogonal to γ , and ζ are emanating from γ . The affine parameter along ζ is x^- . Therefore, x^+ and x^- together form a two-dimensional NFNC.

3.1 x^+ as a parameter

In general, the geodesic equation for a non-affined parameter α is

$$\frac{d^2 x^\mu}{d\alpha^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\alpha} \frac{dx^\sigma}{d\alpha} = f(\alpha) \frac{dx^\mu}{d\alpha}, \quad (70)$$

$$f(\alpha) = - \left(\frac{d^2 \alpha}{d\lambda^2} \right) \left(\frac{d\alpha}{d\lambda} \right)^{-2},$$

with λ affined.

One can choose α to be x^+ , as x^+ is only affine along γ and we are considering geodesics inside the causal diamond. Then the geodesic equations are found to be

$$x^- \mathcal{R} = \left(\frac{d^2 x^+}{d\tau^2} \right) \left(\frac{dx^+}{d\tau} \right)^{-2}, \quad (71)$$

$$\frac{d^2 x^-}{d\alpha^2} + \frac{1}{2} (x^-)^2 \left(2x^- \mathcal{R}^2 + \frac{d\mathcal{R}}{d\alpha} \right) + 2x^- \mathcal{R} \frac{dx^-}{d\alpha} = - \left(\frac{d^2 x^+}{d\tau^2} \right) \left(\frac{dx^+}{d\tau} \right)^{-2} \left(\frac{dx^-}{d\alpha} \right). \quad (72)$$

According to Eqn.(71), Eqn.(72) can be written as

$$\frac{d^2 x^-}{d\alpha^2} + \frac{1}{2} (x^-)^2 \left(2x^- \mathcal{R}^2 + \frac{d\mathcal{R}}{d\alpha} \right) + 3x^- \mathcal{R} \frac{dx^-}{d\alpha} = 0. \quad (73)$$

We expand x^- around $U_F \sim 0$ as follows

$$x^-(\alpha) = \sum_{i=1} f_i(\alpha) U_F^i, \quad (74)$$

where $f_i(\alpha)$ are some functions to be found. The boundary conditions are

$$x^-(0) = 0, \quad (75)$$

$$x^-(V_F) = U_F. \quad (76)$$

Ignoring the higher order term of $\mathcal{R}$ in Eqn.(73), one gets

$$\frac{d^2 x^-}{d\alpha^2} + \frac{1}{2} (x^-)^2 \frac{d\mathcal{R}}{d\alpha} + 3x^- \mathcal{R} \frac{dx^-}{d\alpha} + \mathcal{O}(\mathcal{R}^2) = 0. \quad (77)$$

At the first order, the equation of motion takes the form

$$\frac{d^2 f_1(\alpha)}{d\alpha^2} = 0, \quad (78)$$

It is obvious that

$$f_1(\alpha) = \frac{1}{V_F} \alpha. \quad (79)$$

Then continuing to the second order, one has

$$x^-(\alpha) = \frac{U_F}{V_F} \alpha + U_F^2 f_2(\alpha) + \dots, \quad (80)$$

$$\frac{U_F^2}{2V_F^2} \left(6\alpha \mathcal{R} + \alpha^2 \frac{d\mathcal{R}}{d\alpha} + 2V_F^2 \frac{d^2 f_2}{d\alpha^2} \right) = 0, \quad (81)$$

which gives

$$f_2(\alpha) = C\alpha - \frac{1}{2V_F^2} \int_0^\alpha \left(\int_0^z y(6\mathcal{R}(y) + y \frac{d\mathcal{R}}{dy}) dy \right) dz, \quad (82)$$

where C is a constant and can be fixed by boundary condition (76). It can be found that

$$C = \frac{1}{V_F^3} \int_0^{V_F} \left(\int_0^z y(6\mathcal{R}(y) + y \frac{d\mathcal{R}}{dy}) dy \right) dz. \quad (83)$$

Therefore, x^- can be written as a function of x^+ :

$$x^-(x^+) = x^-(\alpha) = \frac{U_F}{V_F} x^+ + f_2(x^+) U_F^2. \quad (84)$$

3.2 The Solution for τ

Since τ is the proper time, with eqn.(84) we have

$$\begin{aligned} d\tau^2 &= -ds^2 \\ &= 2dx^+ dx^- + (x^-)^2 R_{+-+} (dx^+)^2 \\ &= 2 \left(\frac{U_F}{V_F} + U_F^2 C + U_F^2 \mathfrak{I}(x^+) + \frac{1}{2} \mathcal{R} \frac{U_F^2}{V_F^2} (x^+)^2 + \mathcal{O}(U_F^3, U_F^4) \right) (dx^+)^2, \end{aligned} \quad (85)$$

where

$$\mathfrak{I}(x^+) = -\frac{1}{2V_F^2} \int_0^{x^+} y \left(6\mathcal{R} + y \frac{d\mathcal{R}}{dy} \right) dy.$$

One can see

$$\begin{aligned}
& -\frac{U_F^2}{2V_F^2} \left(\int_0^{x^+} y \left(6\mathcal{R} + y \frac{d\mathcal{R}}{dy} \right) dy - (x^+)^2 \mathcal{R} \right) \\
&= -\frac{U_F^2}{2V_F^2} \left(3y^2 \mathcal{R} \Big|_0^{x^+} - 3 \int_0^{x^+} y^2 \frac{d\mathcal{R}}{dy} dy + \int_0^{x^+} y^2 \frac{d\mathcal{R}}{dy} dy - (x^+)^2 \mathcal{R} \right) \\
&= -\frac{U_F^2}{2V_F^2} \left(2(x^+)^2 \mathcal{R} - 2(x^+)^2 \mathcal{R} + 4 \int_0^{x^+} y \mathcal{R} dy \right) \\
&= -2 \frac{U_F^2}{V_F^2} \int_0^{x^+} y \mathcal{R} dy.
\end{aligned} \tag{86}$$

Thus, Eqn.(85) is simplified and takes the form

$$d\tau^2 = 2 \frac{U_F}{V_F} (1 + U_F V_F C - 2 \frac{U_F}{V_F} \int_0^{x^+} y \mathcal{R} dy) (dx^+)^2,$$

which gives

$$d\tau = \sqrt{\frac{2U_F}{V_F}} \left(1 + U_F V_F C - 2 \frac{U_F}{V_F} \int_0^{x^+} y \mathcal{R} dy \right)^{\frac{1}{2}} dx^+. \tag{87}$$

It will make the calculation more convenient if one expand τ as a function of x^+ around $U_F \sim 0$, that is,

$$\tau(x^+) = \sum_{i=0} U_F^{\frac{i}{2}} g_i(x^+), \tag{88}$$

where $g_i(x^+)$ are some functions to be found. The boundary condition reads

$$\tau(0) = 0. \tag{89}$$

Substituting the expansion into Eqn.(87) at the zeroth order, one finds

$$\frac{dg_0}{dx^+} = 0, \tag{90}$$

which indicates

$$g_0(x^+) = 0. \tag{91}$$

Continuing to $i = 1$, one can see that Eqn.(87) becomes

$$\left(g_1'(x^+) - \sqrt{\frac{2}{V_F}} \right) U_F^{\frac{1}{2}} = 0. \tag{92}$$

Then the solution for $g_1(x^+)$ is simply

$$g_1(x^+) = \sqrt{\frac{2}{V_F}} x^+. \quad (93)$$

As there is no equation to the second order, we can skip to the third order equation.

The equation at the third order is

$$\left(\frac{1}{\sqrt{2}} \left(\frac{1}{V_F} \right)^{\frac{3}{2}} (-CV_F^2 + I(x^+)) + g_3(x^+) \right) U_F^{\frac{3}{2}} = 0, \quad (94)$$

and

$$I(x^+) = \int_0^{x^+} \mathcal{R} y dy. \quad (95)$$

The solution for $g_3(x^+)$ takes the form

$$g_3(x^+) = \frac{1}{\sqrt{2}} \left(C\sqrt{V_F}x^+ - 2 \left(\frac{1}{V_F} \right)^{\frac{3}{2}} \int_0^{x^+} I(z) dz \right). \quad (96)$$

Now we would like to rescale the variables and choose

$$\widetilde{x^+} = \frac{x^+}{V_F}. \quad (97)$$

Similarly, after taking $\tilde{y} = y/V_F$, one can rewrite Eqn.(95) as

$$I(x^+) = \int_0^{x^+} \mathcal{R} y dy = V_F^2 \int_0^{\widetilde{x^+}} \tilde{\mathcal{R}}(\tilde{y}) \tilde{y} d\tilde{y}, \quad (98)$$

and can choose

$$\tilde{I}(\widetilde{x^+}) = \int_0^{\widetilde{x^+}} \tilde{\mathcal{R}}(\tilde{y}) \tilde{y} d\tilde{y}. \quad (99)$$

Therefore, substituting the rescaled results into Eqn.(96) gives

$$g_3(\widetilde{x^+}) = \frac{1}{\sqrt{2}} V_F^{\frac{3}{2}} \left(C\widetilde{x^+} - 2 \int_0^{\widetilde{x^+}} \tilde{I}(\tilde{y}) d\tilde{y} \right). \quad (100)$$

Now back to the expansion (88), we finally find that

$$\begin{aligned} \tau(\widetilde{x^+}) &= \tau(x^+) \\ &= \sqrt{2}(U_F V_F)^{\frac{1}{2}} \widetilde{x^+} + \frac{1}{\sqrt{2}} (U_F V_F)^{\frac{3}{2}} \left(C\widetilde{x^+} - 2 \int_0^{\widetilde{x^+}} \tilde{I}(\tilde{y}) d\tilde{y} \right), \end{aligned} \quad (101)$$

which is rather an interesting result. Note that the tangent vector of γ is T^μ , and that of ζ is S^μ . They are normalised such that

$$g_{\mu\nu}S^\mu T^\nu|_\gamma = -1. \quad (102)$$

Thus, for example, one can make such a choice like ¹

$$x^+ : \quad T^\mu = \frac{1}{\sqrt{2}}(1, 1), \quad (103)$$

$$x^- : \quad S^\mu = \frac{1}{\sqrt{2}}(1, -1). \quad (104)$$

Meanwhile, it is also reasonable to set

$$x^+(A) : \quad T^\mu(A) = \frac{A}{\sqrt{2}}(1, 1), \quad (105)$$

$$x^-(A) : \quad S^\mu(A) = \frac{1}{\sqrt{2}A}(1, -1). \quad (106)$$

It is obvious that $x^+x^- = x^+(A)x^-(A)$, so x^+x^- is invariant of the choice for different geodesics. From Eqn.(101), we know that τ has U_FV_F -dependence, while U_FV_F should be invariant as we illustrated above.

Coming back to Eqn.(101) and taking $\widetilde{x^+} = 1$, one finds the proper length τ' of the timelike geodesic is

$$\begin{aligned} \tau' &= \int_0^{\tau'} d\tau = \tau(\widetilde{x^+} = 1) \\ &= \sqrt{2}\sqrt{U_FV_F} + \frac{1}{\sqrt{2}}(\sqrt{U_FV_F})^3 \left(C - 2 \int_0^1 \tilde{I}(\tilde{y})d\tilde{y} \right) \\ &= \sqrt{2}\sqrt{U_FV_F} + \frac{C_2}{\sqrt{2}}(\sqrt{U_FV_F})^3, \end{aligned} \quad (107)$$

where

$$C_2 = C - 2 \int_0^1 \tilde{I}(\tilde{y})d\tilde{y}. \quad (108)$$

Additionally, we have also used the fact that

$$\int_0^1 d \left(\int_0^{\widetilde{x^+}} \tilde{I}(\tilde{y})d\tilde{y} \right) = \int_0^1 \tilde{I}(\widetilde{x^+})d\widetilde{x^+} = \int_0^1 \tilde{I}(\tilde{y})d\tilde{y}. \quad (109)$$

¹Here we choose the vectors in this metric: $ds^2 = -dt^2 + dx^2$.

From now on we will denote the proper length τ' by τ . With the normalisation (21), one can write

$$R_{+-+-} = R_{\mu\nu\rho\sigma} T^\mu S^\nu T^\rho S^\sigma, \quad (110)$$

and we will illustrate why this is true in Appendix B by showing that the normalisation is consistent with the metric.

Before one writes τ in terms of T^μ and S^ν , it is useful to rescale the variables for constant C (83) as well:

$$\begin{aligned} C &= \frac{1}{V_F^3} \int_0^{V_F} \left(\int_0^z y(6\mathcal{R}(y) + y \frac{d\mathcal{R}}{dy}) dy \right) dz \\ &= \frac{1}{V_F} \int_0^{V_F} \left(\int_0^{\tilde{z}} \tilde{y}(6\tilde{\mathcal{R}}(\tilde{y}) + V_F \tilde{y} \frac{d\tilde{\mathcal{R}}}{d\tilde{y}} \frac{d\tilde{y}}{dy}) d\tilde{y} \right) dz \\ &= \int_0^1 \left(\int_0^{\tilde{z}} \tilde{y}(6\tilde{\mathcal{R}}(\tilde{y}) + \tilde{y} \frac{d\tilde{\mathcal{R}}}{d\tilde{y}}) d\tilde{y} \right) d\tilde{z}. \end{aligned} \quad (111)$$

Additionally, one has

$$\begin{aligned} \int_0^{\tilde{z}} \tilde{y}^2 \frac{d\tilde{\mathcal{R}}}{d\tilde{y}} d\tilde{y} &= \tilde{y}^2 \tilde{\mathcal{R}}(\tilde{y}) \Big|_0^{\tilde{z}} - 2 \int_0^{\tilde{z}} \tilde{\mathcal{R}}(\tilde{y}) \tilde{y} d\tilde{y} \\ &= \tilde{z}^2 \tilde{\mathcal{R}}(\tilde{z}) - 2 \int_0^{\tilde{z}} \tilde{\mathcal{R}}(\tilde{y}) \tilde{y} d\tilde{y}. \end{aligned} \quad (112)$$

As a result, C_2 finally takes the form

$$\begin{aligned} C_2 &= C - 2 \int_0^1 \tilde{I}(\tilde{y}) d\tilde{y} \\ &= \int_0^1 \tilde{\mathcal{R}}(\tilde{z}) \tilde{z}^2 d\tilde{z} + \int_0^1 \left(\int_0^{\tilde{z}} (6\tilde{\mathcal{R}}(\tilde{y}) \tilde{y} - 2\tilde{\mathcal{R}}(\tilde{y}) \tilde{y}) d\tilde{y} \right) d\tilde{z} - 2 \int_0^1 \left(\int_0^{\tilde{z}} (\tilde{\mathcal{R}}(\tilde{y}) \tilde{y}) d\tilde{y} \right) d\tilde{z} \\ &= \int_0^1 \tilde{\mathcal{R}}(\tilde{z}) \tilde{z}^2 d\tilde{z} + \int_0^1 \left(\int_0^{\tilde{z}} 2\tilde{\mathcal{R}}(\tilde{y}) \tilde{y} d\tilde{y} \right) d\tilde{z}. \end{aligned} \quad (113)$$

This gives that

$$\begin{aligned} \tau &= \sqrt{2} \sqrt{U_F V_F} + \frac{(\sqrt{U_F V_F})^3}{\sqrt{2}} \left(\int_0^1 \tilde{z}^2 \tilde{R}_{\mu\nu\rho\sigma} T^\mu S^\nu T^\rho S^\sigma d\tilde{z} \right. \\ &\quad \left. + \int_0^1 \left(\int_0^{\tilde{z}} (2\tilde{R}_{\mu\nu\rho\sigma} T^\mu S^\nu T^\rho S^\sigma \tilde{y}) d\tilde{y} \right) d\tilde{z} \right). \end{aligned} \quad (114)$$

Therefore, we have found the relationship between τ and $U_F V_F$ in a ‘non-coincident limit’.

3.3 The Two-volume of a Causal Diamond

To get the volume of the causal diamond, one can use formula (1). It is obvious that in metric (38), $|g| = 1$, so the volume is

$$V(x, y) = \int_0^{V_F} dx^+ \int_0^{\sigma(x^+)} dx^- = \int_0^{V_F} (x^-|_{\sigma}) dx^+, \quad (115)$$

where $\sigma(x^+)$ is the upper limit of x^- as a function of x^+ . As shown in Fig.7, since we are studying a causal diamond in a general two-dimensional metric, the upper limit of x^- is not simply U_F .

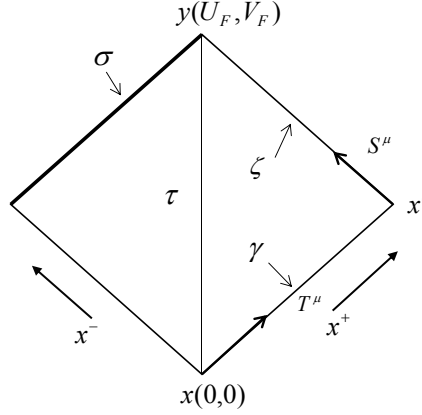


Figure 7: Here σ is a null geodesic and we would like to calculate the 2-volume by integrating along σ . In flat spacetime, σ is a simple straight line $x^- = U_F$. However, in general 2-dimensional spacetime, it is not necessary a straight line and in fact should be a function of x^+ .

Similarly, σ is also a null geodesic like γ and ζ . Thus, with the null condition, one has

$$\frac{dx^+}{d\alpha} \left(-2 \frac{dx^-}{d\alpha} - (x^-)^2 \mathcal{R}(x^+) \frac{dx^+}{d\alpha} \right) = 0, \quad (116)$$

where α is the affine parameter along σ . This implies that

$$2 \frac{dx^-}{dx^+} + (x^-)^2 \mathcal{R}(x^+) = 0, \quad (117)$$

which gives

$$x^-(x^+)|_\sigma = \frac{1}{-C' + \frac{1}{2} \int_0^{x^+} \mathcal{R}(y) dy}. \quad (118)$$

Using the boundary condition that $x^-(V_F) = U_F$, one finds

$$C' = -\frac{1}{U_F} + \frac{1}{2} \int_0^{V_F} \mathcal{R}(y) dy. \quad (119)$$

It is not convenient to get the volume if simply substituting Eqn.(118) into Eqn.(115), so we would first expand Eqn.(118) around $U_F \sim 0$:

$$x^-|_\sigma = U_F + \frac{1}{2} \left(-\int_0^{x^+} \mathcal{R}(y) dy + \int_0^{V_F} \mathcal{R}(y) dy \right) U_F^2. \quad (120)$$

Hence, the volume is

$$V(x, y) = \int_0^{V_F} \left(U_F + \frac{1}{2} \left(-\int_0^{x^+} \mathcal{R}(y) dy + \int_0^{V_F} \mathcal{R}(y) dy \right) U_F^2 \right) dx^+. \quad (121)$$

Rescaling variables as what has been done above, i.e. taking $\tilde{y} = y/V_F$ and $\widetilde{x^+} = x^+/V_F$, one finds

$$\begin{aligned} \frac{1}{2} \int_0^{V_F} \left(\int_0^{x^+} \mathcal{R}(y) dy \right) dx^+ &= \frac{V_F}{2} \int_0^{V_F} \left(\int_0^{\widetilde{x^+}} \tilde{\mathcal{R}}(\tilde{y}) d\tilde{y} \right) dx^+ \\ &= \frac{V_F}{2} \int_0^{V_F} \tilde{I}_2(\widetilde{x^+}) d\widetilde{x^+} = \frac{V_F^2}{2} \int_0^1 \tilde{I}_2(\widetilde{x^+}) d\widetilde{x^+} = S \cdot V_F^2, \end{aligned} \quad (122)$$

and

$$\tilde{I}_2(\widetilde{x^+}) = \int_0^{\widetilde{x^+}} \tilde{\mathcal{R}}(\tilde{y}) d\tilde{y}, \quad (123)$$

$$S = \frac{1}{2} \int_0^1 \tilde{I}_2(\widetilde{x^+}) d\widetilde{x^+} = \text{const.} \quad (124)$$

Next, one has

$$\frac{1}{2} \int_0^{V_F} \mathcal{R}(y) dy = \frac{1}{2} V_F \int_0^1 \tilde{\mathcal{R}}(\tilde{y}) d\tilde{y} = V_F \cdot Y, \quad (125)$$

where

$$Y = \frac{1}{2} \int_0^1 \tilde{\mathcal{R}}(\tilde{y}) d\tilde{y} = \text{const.} \quad (126)$$

Now the volume (121) is

$$V(x, y) = U_F V_F + U_F^2 V_F^2 (Y - S) + \mathcal{O}(U_F^3 V_F^3), \quad (127)$$

which is exactly the result in metric (35). In fact, one can check that both Eqn.(84) and Eqn.(127) are still true even if the metric for NFNC is extended to higher orders of x^- according to Eqn.(34) and we will prove it later. If we now compare what we have found with Eqn.(6), something interesting happens. Eqn.(127) shows that the two-volume has $U_F V_F$ -dependence, whilst $U_F V_F$ keeps invariant because of the normalisation. However from Eqn.(6) one finds that in four-dimensional spacetime, the volume does not necessarily depend on $U_F V_F$. In fact, in four-dimension the situation is rather complicated, since the timelike geodesic may not simply lie on a two-plane.

The next thing to do is to find the relationship between $V(x, y)$ and the proper length τ . In order to obtain it, one should first write $U_F V_F$ as a function of τ . It gives a rather complicated result if one solves the Eqn.(107) directly, so it is necessary to use another method as follows. Assume that

$$z = \sqrt{U_F V_F} = d_0 + d_1 \tau + d_2 \tau^2 + d_3 \tau^3 + \mathcal{O}(\tau^4), \quad (128)$$

where d_i are a set of constants. Substituting Eqn.(128) into (107) gives

$$\begin{aligned} \tau = & \left(\sqrt{2}d_0 + \frac{C_2}{\sqrt{2}}d_0^3 \right) + \left(\sqrt{2}d_1 + \frac{3C_2}{\sqrt{2}}d_0^2 d_1 \right) \tau \\ & + \left(\frac{3C_2}{\sqrt{2}}d_0 d_1^2 + \sqrt{2}d_2 + \frac{3C_2}{\sqrt{2}}d_0^2 d_2 \right) \tau^2 \\ & \left(\frac{C_2}{\sqrt{2}}d_1^3 + 3\sqrt{2}C_2 d_0 d_1 d_2 + \sqrt{2}d_3 + \frac{3C_2}{\sqrt{2}}d_0^2 d_3 \right) \tau^3 + \mathcal{O}(\tau^4). \end{aligned} \quad (129)$$

Therefore,

$$\begin{aligned} d_0 &= 0, \\ d_1 &= \frac{1}{\sqrt{2}}, \\ d_2 &= 0, \\ d_3 &= -\frac{C_2}{4\sqrt{2}}. \end{aligned}$$

Thus,

$$\sqrt{U_F V_F} = \frac{\tau}{\sqrt{2}} - \frac{C_2 \tau^3}{4\sqrt{2}}. \quad (130)$$

Insert it into Eqn.(127), then

$$\begin{aligned}
V(x, y) &= \frac{1}{2}\tau^2 + \frac{1}{4}(Y - S - C_2)\tau^4 \\
&= \frac{1}{2}\tau^2 + \frac{1}{4} \left(\int_0^1 d\tilde{z} \tilde{\mathcal{R}}(\tilde{z}) \left(\frac{1}{2} - \tilde{z}^2 \right) \right. \\
&\quad \left. - \int_0^1 d\tilde{z} \int_0^{\tilde{z}} d\tilde{y} \tilde{\mathcal{R}}(\tilde{y}) \left(\frac{1}{2} + 2\tilde{y} \right) \right) \tau^4 + \mathcal{O}(\tau^6). \tag{131}
\end{aligned}$$

Finally, substituting Eqn.(110) into Eqn.(131), one gets

$$\begin{aligned}
V(x, y) &= \frac{1}{2}\tau^2 + \frac{1}{4} \left(\int_0^1 d\tilde{z} \tilde{R}_{\mu\nu\rho\sigma}(\tilde{z}) T^\mu S^\nu T^\rho S^\sigma \left(\frac{1}{2} - \tilde{z}^2 \right) \right. \\
&\quad \left. - \int_0^1 d\tilde{z} \int_0^{\tilde{z}} d\tilde{y} \tilde{R}_{\mu\nu\rho\sigma}(\tilde{y}) T^\mu S^\nu T^\rho S^\sigma \left(\frac{1}{2} + 2\tilde{y} \right) \right) \tau^4 + \mathcal{O}(\tau^6), \tag{132}
\end{aligned}$$

which is the relation we were looking for.

3.4 Results Verification

Now we would like to calculate the two-volume in metric (35) and see whether the result is in consistent with Eqn.(127). Using formula (1) and letting $V_2(x, y)$ be the volume in this metric, obtains [32]

$$\begin{aligned}
V_2(x, y) &= \int_{diamond} \sqrt{|g|} = \int_0^U \int_0^V (1 - a(v)u) du dv \\
&= UV - \frac{1}{2}U^2 \int_0^V a(v) dv. \tag{133}
\end{aligned}$$

Additionally, one can give the explicit form of V_2 in terms of the curvature and the tangent vectors [32]. First note that the Riemann tensor for metric (35) is

$$R_{uvuv} = \frac{1}{2} \partial_v a(v) + \mathcal{O}(u). \tag{134}$$

Next integration by parts gives

$$\int_0^V a(v) dv = \int_0^V (V - v) \partial_v a(v) dv. \tag{135}$$

Then the volume(133) can be rewritten as

$$\begin{aligned}
V_2(x, y) &= UV - \int_0^V R_{uvuv} U^2 (V - v) dv \\
&= UV - \int_0^V R_{abcd} (x - x')^a (x - x')^c (x - x'')^b dx''^b, \tag{136}
\end{aligned}$$

where x'' is a point on geodesic γ between x and x' .

Now according to Eqn.(49) and Eqn.(50), one gets

$$(U_F, V_F) = (U - \frac{1}{2}a(V)U^2, V). \quad (137)$$

Additionally,

$$U_F^2 = U^2 + \mathcal{O}(U^3), \quad (138)$$

$$U_F V_F = UV - \frac{1}{2}U^2 V a(V). \quad (139)$$

Then from Eqn.(66), Eqn.(124) and Eqn.(126), it can be shown that

$$\begin{aligned} Y &= \frac{1}{2} \int_0^1 \tilde{\mathcal{R}}(\tilde{y}) d\tilde{y} = \frac{1}{2V_F} \int_0^{V_F} \mathcal{R}(y) dy \\ &= \frac{1}{2V} \int_0^V a'(z) dz \\ &= \frac{1}{2V} a(V), \end{aligned} \quad (140)$$

and

$$\begin{aligned} S &= \frac{1}{2} \int_0^1 \left(\int_0^{\tilde{x}^+} \tilde{\mathcal{R}}(\tilde{y}) d\tilde{y} \right) d\tilde{x}^+ \\ &= \frac{1}{2V_F^2} \int_0^{V_F} \left(\int_0^{x^+} a'(z) dz \right) dx^+ \\ &= \frac{1}{2V^2} \int_0^V a(v) dv. \end{aligned} \quad (141)$$

Finally comparing Eqn.(127) with Eqn.(133), we can see

$$\begin{aligned} V(x, y) &= U_F V_F + U^2 V_F^2 (Y - S) \\ &= U_F V_F + \frac{1}{2} U_F^2 V_F a(V) - \frac{1}{2} U_F^2 \int_0^V a(v) dv \\ &= UV - \frac{1}{2} U^2 V a(V) + \frac{1}{2} U^2 V a(V) - \frac{1}{2} U^2 \int_0^V a(v) dv \\ &= UV - \frac{1}{2} U^2 \int_0^V a(v) dv \\ &= V_2(x, y). \end{aligned} \quad (142)$$

Hence the expression of two-volume in null coordinates and NFNC are in consistent with each other, which is what we have expected.

4 Results for the Metric with Higher Order Terms

To show that Eqn.(84) and Eqn.(127) remain true for the NFNC metric in higher order, one only needs to show that the extension of the metric will not bring new terms containing U_F^2 into the solution for $x^-(x^+)$.

First write the metric to the third order of x^- :

$$ds^2 = -2dx^+dx^- - (x^-)^2 R_{+--+}(dx^+)^2 - \frac{1}{3} R_{+--+;-}(x^-)^3(dx^+)^2 + \mathcal{O}((x^-)^4). \quad (143)$$

Following the steps from Eqn.(70) to Eqn.(73), one gets the equation of motion

$$\begin{aligned} 0 = & \frac{d^2 x^-}{(dx^+)^2} + \frac{1}{2} x^- \frac{dx^-}{dx^+} \left(6R_{+--+} + x^- (3R_{+--+;-} + x^- \frac{dR_{+--+;-}}{dx^-}) \right) \\ & + \frac{1}{18} (x^-)^2 \left(9R'_{+--+}(x^+) + (x^-)^4 R_{+--+;-} \frac{dR_{+--+;-}}{dx^-} \right. \\ & \left. + 3(x^-)^2 R_{+--+} (5R_{+--+;-} + x^- \frac{dR_{+--+;-}}{dx^-}) + 3x^- \frac{dR_{+--+;-}}{dx^+} \right) + \dots \end{aligned} \quad (144)$$

Same as what we have done before, expand x^- as

$$x^-(x^+) = \sum_{i=1} f_i(x^+) U_F^i. \quad (145)$$

To the first order:

$$x^- = U_F f_1(x^+), \quad (146)$$

and the equation of motion is

$$f_1''(x^-) U_F + (O)(U_F^2) = 0. \quad (147)$$

So

$$f_1(x^+) = \frac{1}{V_F} x^+. \quad (148)$$

Then

$$x^- = \frac{U_F}{V_F} x^+ + U_F^2 f_2(x^+), \quad (149)$$

and the equation is

$$\frac{U_F^2}{2V_F^2} \left(6\alpha \mathcal{R} + \alpha^2 \frac{d\mathcal{R}}{d\alpha} + 2V_F^2 \frac{d^2 f_2}{d\alpha^2} \right) + \mathcal{O}(U_F^3) = 0, \quad (150)$$

which is exactly the same as the equation of motion (81). Thus, Eqn.(84) remains true in higher order NFNC metric.

Then let us check whether Eqn.(127) is still true. As we see, the only possible contribution comes from $\sqrt{|g|}$, since terms of $(x^-)^2$ from the metric will be ignored in Eqn.(116). However, from the metric (143), we can see the new term is the $++$ -component of the metric, which is on the diagonal and does not contribute to $|g|$ as the $---$ -component is still zero. We can even continue to the forth order where

$$\begin{aligned}
ds^2 = & -2dx^+dx^- - (x^-)^2 R_{+-+-}(dx^+)^2 - \frac{1}{3} R_{+-+-;-}(x^-)^3(dx^+)^2 \\
& + (\frac{1}{3} R_{+-+-} R^+{}_{--} - \frac{1}{12} R_{+-+-;--})(x^-)^4(dx^+)^2.
\end{aligned} \tag{151}$$

Obviously, nothing contributes to $---$ -components and $|g|$ is still 1. In conclusion, both Eqn.(84) and Eqn.(127) are true even if one works in a higher order metric. Consequently, the formula (132) still applies in the metric with higher order terms.

5 An extension to Four-dimensional spacetime

Therefore, we have got the volume of a causal diamond in two-dimensional spacetime, and now we would like to make an extension to four-dimensional spacetime. According to 't Hooft's formula (2), the normalisation (21), and our result in two-dimensional spacetime (132), it is reasonable to assume that in four-dimensional spacetime the volume reads

$$V(x, y) = \frac{\pi}{24} \tau^4 + X \tau^6 + \dots, \quad (152)$$

where

$$\begin{aligned} X = & \int_0^1 d\tilde{z} \int_0^{\tilde{z}} d\tilde{y} (a + b\tilde{y}) (R + cR_{\mu\nu}T^\mu S^\nu + dR_{\mu\nu\rho\sigma}T^\mu S^\nu T^\rho S^\sigma) \\ & + \int_0^1 d\tilde{z} (f + g\tilde{z} + h\tilde{z}^2) (R + jR_{\mu\nu}T^\mu S^\nu + kR_{\mu\nu\rho\sigma}T^\mu S^\nu T^\rho S^\sigma). \end{aligned} \quad (153)$$

Here, R , $R_{\mu\nu}$ and $R_{\mu\nu\rho\sigma}$ are Ricci scalar, Ricci tensors and Riemann tensors respectively.

To fix a, b and other coefficients, one has to calculate the volume in nine different spacetimes (because we have nine coefficients), just like what has been done in [11]. The exact expression of the four-volume of a long skinny interval may be different from Eqn.(153) as we may have missed some higher order term of $\tilde{y}$ or $\tilde{z}$. But this problem can be solved by an explicit computation. Unfortunately, as the calculation could be quite complicated, we are unable to finish it in this paper.

6 Conclusions and discussions

In this paper, we calculated the volume of a ‘long skinny interval’ in general two-dimensional spacetime and gave the form it may take when extended to four-dimensional spacetime. It is interesting to compare our results with those of [11]. In [11], the volume of a ‘short fat interval’ is calculated. There are two points x and y , and x is in the chronological past of y , as shown in Fig.8. The construction of a ‘short fat interval’ ends at a point o .

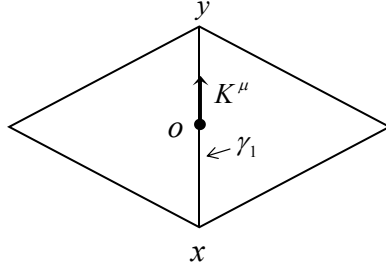


Figure 8: The construction of a ‘short fat interval’ is centred at the point o , the mid point of the timelike geodesic γ_1 . The point x and y are connected via γ_1 , whose proper length is τ . When x and y are close enough, the volume of a causal diamond will depend on the curvature at o .

Suppose exists a unique timelike geodesic γ_1 whose proper length is τ . Point o is the mid point of γ_1 with $t = 0$, where t is the affine parameter along γ_1 . The tangent vector along γ_1 is K^μ . Then in a general curved spacetime, assume that x and y are getting sufficiently close, the volume of this causal diamond will be [11, 15]

$$V(x, y) = \frac{\pi}{24} \tau^4 \left(1 - \frac{1}{180} \tau^2 R(o) + \frac{1}{30} R_{\mu\nu}(o) T^\mu T^\nu + \dots \right). \quad (154)$$

So the corrections for the volume of a ‘short fat interval’ depend on the curvature at point o .

Comparing Eqn.(154) with our result (131), one finds that the volume of a ‘long skinny interval’ does not depend on the curvature at one single point, but the curvature along the whole geodesic γ_2 , which is plotted in Fig.9.

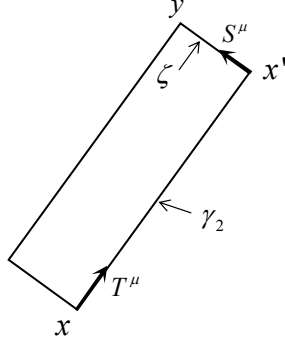


Figure 9: Unlike the ‘short fat interval’, a ‘long skinny interval’ is constructed by two null geodesics, γ_2 and ζ . Also, the timelike geodesic from x and y has a proper length of τ . If the interval between y and x' is infinitely small, then the volume of such causal diamond will involve the curvature along the whole geodesic γ_2 , not a single point, since x' is fixed.

This is mainly because the construction of a ‘long skinny interval’ involves two null geodesics, γ_2 and ζ . As x' is fixed, certainly there will be a dependence on the curvature along γ_2 . In addition, since both γ_2 and ζ are involved, it is reasonable to suppose that their tangent vectors, T^μ and S^μ can contribute to the volume, which can be seen from Eqn.(153).

With the volume of a causal diamond, one is now enabled to calculate the integration (20), studying deeper into a discrete D’Alembertian and therefore the theory of causal sets. However, our result is limited since we did not find the behaviour of the volume of a causal diamond in four or higher dimensions and it will be our interest in future.

Acknowledgements

The author would like to thank Professor Fay Dowker, without whom the paper cannot be finished. Thank you for introducing such a fascinating topic and offering great support all the time.

The author would also like to thank S. N. Solodukhin and G. W. Gibbons for the useful discussions at the first stage of this work.

Finally, the author would like to thank Szymon Brych, who helped with revising this paper.

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A Affine Parameter in Four-dimensional Spacetime

It is essential to find the affine parameter along a null geodesic before one constructs a null Fermi normal coordinate. Here we consider a conformally flat 4-dimensional spacetime, and the metric is

$$ds^2 = f(x^\mu)(-dt^2 + dx^2 + dy^2 + dz^2), \quad (\text{A.1})$$

where $x^\mu = (t, x, y, z)$. Then using geodesic equation, one can find the affine parameter λ along a null geodesic. The geodesic equation can be written as

$$\frac{d}{d\lambda} \left(g_{\alpha\mu} \frac{dx^\mu}{d\lambda} \right) = \frac{1}{2} (\partial_\alpha g_{\rho\sigma}) \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda}, \quad (\text{A.2})$$

and the null geodesic condition is

$$-\left(\frac{dt}{d\lambda}\right)^2 + \left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2 + \left(\frac{dz}{d\lambda}\right)^2 = 0. \quad (\text{A.3})$$

Hence one has

$$\frac{d}{d\lambda} \left(f(x^\mu) \frac{dx^\mu}{d\lambda} \right) = 0, \quad (\text{A.4})$$

which implies

$$f(x^\mu) \frac{dx^\mu}{d\lambda} = \text{const}. \quad (\text{A.5})$$

If we suppose $x^\mu = N^\mu q + x_0^\mu$, where N^μ and x_0^μ are some four-vectors and q is a function of λ , then finally we find

$$f(x^\mu) \frac{dq}{d\lambda} = \text{const}. \quad (\text{A.6})$$

B Consistency of Normalisation and the Metric

In this section we would like to show the consistency of our normalisation (21) with the metric (34) to the zeroth order:

$$ds^2 = -2dx^+dx^-. \quad (\text{B.1})$$

The most general metric for two-dimensional coordinates $x^\mu = \{t, x\}$ is

$$ds^2 = -dt^2 + dx^2, \quad (\text{B.2})$$

and

$$T^\mu = \frac{dx^\mu}{dx^+}, \quad (\text{B.3})$$

since T^μ is the tangent vector for geodesic γ and x^+ is the affine parameter along γ .

Similarly,

$$S^\mu = \frac{dx^\mu}{dx^-}. \quad (\text{B.4})$$

Because of the normalisation, one can make a choice of T^μ and S^μ like what we have done above:

$$T^\mu = \frac{1}{\sqrt{2}}(1, 1), \quad (\text{B.5})$$

$$S^\mu = \frac{1}{\sqrt{2}}(1, -1). \quad (\text{B.6})$$

Thus, one finds

$$\frac{dt}{dx^+} = \frac{dx}{dx^+} = \frac{1}{\sqrt{2}}, \quad (\text{B.7})$$

as well as

$$\frac{dt}{dx^-} = -\frac{dx}{dx^-} = \frac{1}{\sqrt{2}}. \quad (\text{B.8})$$

Therefore,

$$t = \frac{1}{\sqrt{2}}(x^+ + x^-), \quad (\text{B.9})$$

$$x = \frac{1}{\sqrt{2}}(x^+ - x^-). \quad (\text{B.10})$$

Finally, one obtains

$$\begin{aligned} ds^2 &= -dt^2 + dx^2 \\ &= -\frac{1}{2} \left((dx^+)^2 + (dx^-)^2 + 2dx^+dx^- \right) + \frac{1}{2} \left((dx^+)^2 + (dx^-)^2 - 2dx^+dx^- \right) \\ &= -2dx^+dx^-. \end{aligned} \tag{B.11}$$

C Conformally Flat Two-dimensional Spacetime

Any tow-dimensional Riemannian manifold $\{\mathcal{M}, g\}$ is conformally flat. Here is the derivation[33]:

Let $\{x, y\}$ be the original local coordinates and the metric is

$$ds^2 = g_{xx}dx^2 + 2g_{xy}dxdy + g_{yy}dy^2. \quad (\text{C.1})$$

The metric can be written as

$$ds^2 = \left(\sqrt{g_{xx}}dx + \frac{g_{xy} + i\sqrt{g}}{\sqrt{g_{xx}}}dy \right) \left(\sqrt{g_{xx}}dx + \frac{g_{xy} - i\sqrt{g}}{\sqrt{g_{xx}}}dy \right), \quad (\text{C.2})$$

where

$$g = g_{xx}g_{yy} - g_{xy}^2. \quad (\text{C.3})$$

Suppose there exists a factor

$$\lambda(x, y) = m_1(x, y) + im_2(x, y) \quad (\text{C.4})$$

satisfying that

$$\lambda \left(\sqrt{g_{xx}}dx + \frac{g_{xy} + i\sqrt{g}}{\sqrt{g_{xx}}}dy \right) = du + idv, \quad (\text{C.5})$$

$$\bar{\lambda} \left(\sqrt{g_{xx}}dx + \frac{g_{xy} - i\sqrt{g}}{\sqrt{g_{xx}}}dy \right) = du - idv. \quad (\text{C.6})$$

Then

$$ds^2 = \frac{1}{|\lambda|^2}(du^2 + dv^2). \quad (\text{C.7})$$

Therefore, any two-dimensional Riemannian manifold is conformally flat.